AWS Cloud Institute: Cloud Operations 2 Syllabus

Course overview

In this course, you will work with application deployment services. You will learn about DevOps team enablement, the AWS Cloud Development Kit, and Infrastructure as Code and its benefits. This course covers distributed tracing, troubleshooting, and open-source monitoring.

Instructor-led training (ILT) sessions

Attendance at ILT sessions is strongly encouraged, but attendance does not count toward course completion. You can use the ILT sessions to explore concepts presented in the online learning content in greater depth, connect with AWS instructors and ACI learners, and prepare for the weekly final assessment. All ILT sessions, except for office hours, are recorded and will be available on-demand for you to watch when you have time. Multiple live ILT sessions are offered each day to accommodate a range of schedules.

Course completion requirements

At the end of each module, you will complete a multiple-choice final assessment. You have unlimited opportunities to achieve a passing score on each final assessment. You must complete the final assessments with a score of 80% or higher to receive credit for the course. ILT attendance or watching ILT recordings is strongly encouraged but is not required for course completion.

Week 1: DevOps 2 – Part 1

Goal

Build your knowledge of continuous integration and continuous delivery (CI/CD) by exploring its significance and best practices. Learn how to automate testing processes with continuous integration and AWS CodeBuild to make sure there are efficient and reliable software-delivery pipelines. Gain insight about integrating code reviews into automated testing workflows to foster collaboration and code quality.

Learning objectives

- Define and explain the core principles of CI/CD
- Describe the benefits and challenges of implementing CI/CD
- Identify the key components of a CI/CD pipeline
- Explain best practices for a CI/CD pipeline
- Explain the benefits of automating testing code changes
- Describe code coverage and its purpose for code testing
- Identify non-AWS testing code tools
- Describe how CodeBuild performs application testing
- State the types of testing CodeBuild can perform
- Configure CodeBuild to perform application testing
- Describe the relationship between automated testing and code reviews
- Explore techniques for automated testing and code review integration
- Explain what Amazon CodeGuru is, its benefits, and use cases
- Understand how CodeGuru helps with manual code reviews

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Outline

1. **Review of Continuous Integration and Continuous Delivery**
    a. **Review of Continuous Integration and Continuous Delivery (CI/CD)**
        i. Review of DevOps Principles
        ii. CI/CD Core Principles
        iii. Benefits and Challenges of CI/CD
        iv. Best Practices for CI/CD Pipelines
        v. AWS SimuLearn Challenge: CI/CD Pipelines for APIs
        vi. Knowledge Check
2. **Automate Testing with Continuous Integration**
    a. **Automate Testing with Continuous Integration**
        i. Testing Code Changes
        ii. Unit Testing in CI/CD
        iii. Benefits of Automating Testing Code Changes
        iv. Code Coverage and its Purpose for Code Testing
        v. Non-AWS Unit Testing Code Tools
        vi. Knowledge Check
3. **Automate Testing with AWS CodeBuild**
    a. **Automate Testing with AWS CodeBuild**
        i. Review of AWS CodeBuild
        ii. Types of Testing in AWS CodeBuild
        iii. Lab: Using AWS CodePipeline for Unit Testing
        iv. Knowledge Check
4. **Integrating Code Reviews in Automated Testing**
    a. **Integrating Code Reviews in Automated Testing**
        i. Automated Testing and Code Reviews
        ii. Methods for Automated Testing and Code Review Integration
        iii. Amazon CodeGuru
        iv. Lab: Automating Code Reviews with Amazon CodeGuru
        v. Knowledge Check

Hands-on labs

- **AWS SimuLearn Challenge: CI/CD Pipelines for APIs (60 min)**
  In this lab, you are assigned the vital task of repairing and finalizing the CI/CD pipeline for the company's APIs. Although the core architecture is already in place, some key configurations are missing. Your goal is to complete the required tests and make sure the pipeline is fully operational. When your task is successfully completed, the Task Manager application will be fully functional, so employees can create and manage tasks efficiently.
- **Lab: Using AWS Code Pipeline for Unit Testing (60 min)**
  In this lab, you will use AWS CodePipeline to perform unit testing for an application.
- **Lab: Automating Code Reviews with Amazon CodeGuru (60 min)**
  In this lab, you will use CodeGuru to perform a review on existing code.

Week 2: DevOps 2 – Part 2

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Goal

Learn about the CI/CD pipeline and how it enables seamless automated software delivery. You will explore various deployment strategies for CI/CD to help ensure efficient and reliable releases. You will also learn how to provide consistent and dependable DevOps environments to enable streamlined development and operations processes.

Learning objectives

- Explain an application developer's role and responsibilities with continuous delivery
- Explain how CodePipeline works and its benefits
- Describe AWS CodeDeploy and its benefits and purpose
- Explain CodeDeploy components
- Explain the need for a deployment strategy in continuous delivery and deployment
- Describe various deployment strategies
- Evaluate application deployment requirements
- Recommend an appropriate deployment strategy
- Describe the need for managing DevOps standardization
- Explain the requirements of DevOps standardization tools
- Describe AWS Proton, its purpose, and the benefits
- Create services and environments with AWS Proton
- Explain how AWS Service Catalog provides a self-service experience

Outline

1. **Integrating with the CI/CD Pipeline**
    a. **Integrating with the CI/CD Pipeline**
        i. The Application Developer Role in CI/CD
        ii. Continuous Delivery Pipeline Requirements
        iii. Review of AWS CodeDeploy
        iv. Review of AWS CodePipeline
        v. Code Testing in Continuous Delivery
        vi. Lab: Using AWS CodePipeline for Integration Testing
        vii. Knowledge Check
2. **Deployment Strategies for Continuous Delivery and Deployment**
    a. **Deployment Strategies for Continuous Delivery and Deployment**
        i. Types of Deployment Strategies for CD
        ii. In-Place Deployments
        iii. Immutable Updates
        iv. Rolling Deployments
        v. Blue/Green Deployments
        vi. Lab: Blue/Green Deployments with AWS CodeDeploy and Amazon EC2
        vii. Knowledge Check
3. **Providing Reliable and Consistent DevOps Environments**
    a. **Providing Reliable and Consistent DevOps Environments**
        i. Managing DevOps Standards
        ii. AWS Proton
        iii. Demo: Creating Environments and Services with AWS Proton
        iv. AWS Service Catalog
        v. Demo: Create Deployment Pipeline with AWS Service Catalog

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

vi. Lab: Deploying Standardized Assets Using AWS Service Catalog
vii. Knowledge Check

Hands-on labs

- **Lab: Using AWS CodePipeline for Integration Testing (60 min)**
  This lab focuses on integrating automated integration testing into a Continuous Integration and Continuous Delivery (CI/CD) pipeline using CodePipeline. You will gain hands-on experience in configuring the pipeline to include an integration testing stage, which validates the application's functionality before deployment.
- **Lab: Blue/Green Deployments with AWS CodeDeploy and Amazon EC2 (90 min)**
  In this lab, you will set up a blue-green deployment strategy using CodeDeploy for an existing AWS CodeCommit repository and CodePipeline deployment. You will learn how to create a new deployment group, add an application load balancer, and modify the existing deployment to perform blue-green deployments across the two Amazon Elastic Compute Cloud (Amazon EC2) instances.
- **Lab: Deploying Standardized Assets Using AWS Service Catalog (90 min)**
  In this lab, you will create an AWS Service Catalog portfolio that contains four products. Each Service Catalog product is backed by an AWS CloudFormation template, which is supplied as part of the lab.

Week 3: DevOps 2 – Part 3

Goal

Explore the concept of development, security, and operations (DevSecOps) to understand its significance in integrating security practices throughout the software development lifecycle. You will also learn about practical approaches to applying DevSecOps principles.

Learning objectives

- Describe DevSecOps and its role in securing software delivery
- Identify DevSecOps components
- Identify common vulnerabilities in software development
- Explain common security testing practices in software development
- Identify AWS and third-party security testing tools
- Explain the role of CodeGuru Security in security testing
- Explain how Amazon Inspector automates vulnerability management at scale
- Explain DevSecOps best practices

Outline

1. **DevSecOps**
   a. **DevSecOps**
      i. The Role of DevSecOps
      ii. Key Components of DevSecOps
      iii. Common Vulnerabilities in Software Development
      iv. Security Testing Practices in DevSecOps
      v. Demo: SAST
      vi. Knowledge Check
2. **Applying DevSecOps**
   a. **Applying DevSecOps**
      i. Security Testing Tools

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

ii. Getting Started with Amazon CodeGuru Security
iii. Getting Started with Amazon Inspector
iv. Automating Vulnerability Management with Amazon Inspector
v. Scenario: DevSecOps Best Practices
vi. Lab: Vulnerability Scanning with Amazon Inspector
vii. AWS SimuLearn: Secure Self-Service Infrastructure
viii. AWS SimuLearn: Securing Your Servers
ix. Knowledge Check

Hands-on labs

- **Lab: Using Amazon Inspector for Vulnerability Scanning (75 min)**
  In this lab, you will use Amazon Inspector to scan for vulnerabilities in your AWS resources, specifically Amazon EC2 instances and AWS Lambda functions. You learn how to configure Amazon Inspector, interpret the vulnerability reports, and remediate the findings.
- **AWS SimuLearn Lab: Secure Self-Service Infrastructure (60 min)**
  In this lab, you will help the head of the IT Services department at a web gaming company who has limited staff to deploy IT services to other departments. The IT Services department is looking for a way to hand off these deployments to the other departments by implementing a self-service tool in which configurations already meet compliance requirements and resources are secure.
- **AWS SimuLearn Lab: Securing Your Servers (60 min)**
  In this lab, you will help a mining operation that built an enterprise resource planning (ERP) application to process job applicant submissions. The application is deployed and running, but the mining operation wants help setting least privilege permissions to limit the application's effect on AWS resources.

Week 4: Infrastructure as Code – Part 1

Goal

Build on what you have learned about infrastructure as code (IaC) and how it fits into a DevOps workflow. Discover both AWS and Non-AWS IaC tools, such as AWS CloudFormation, AWS CDK, and Terraform.

Learning objectives

- Define IaC and its benefits
- Describe the problems IaC solves in traditional infrastructure management
- Identify the benefits of IaC for application developers in a DevOps environment
- Explain how IaC fits into the DevOps workflow
- Identify an application developer’s responsibilities in writing and maintaining IaC templates
- Describe how application developers and infrastructure teams work together over IaC
- Explain the common components and requirements of an IaC tool
- Describe both AWS and non-AWS IaC tools
- Describe Rules in CloudFormation and discuss working with them
- Explain Intrinsic functions in CloudFormation and discuss working with them
- Describe Conditions in CloudFormation and discuss working with them
- Define the *DependsOn* attribute and describe its use
- Define Template Macros and describe how they work
- Describe the benefits of modules and explain how they work
- Explain using the CloudFormation Designer

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Outline

1. **Review Infrastructure as Code**
    a. **Review of Infrastructure as Code**
        i. Define Infrastructure as code
        ii. Knowledge Check
2. **IAC as a Part of a DevOps Workflow**
    a. **IaC as a Part of a DevOps Workflow**
        i. IaC as a Part of a DevOps Workflow
        ii. Lab: Deploying Infrastructure as Code with AWS CodePipeline
        iii. Developer Responsibilities in Writing and Maintaining IaC Templates
        iv. How Developers and Infrastructure Teams Work Together over IaC
        v. Knowledge Check
3. **IaC Tools**
    a. **IaC Tools**
        i. Common Components and Requirements of an IaC Tool
        ii. AWS and Third-Party IaC Tools
        iii. Lab: Building a Serverless API with Infrastructure Composer
        iv. Knowledge Check
4. **Building Templates in AWS CloudFormaton**
    a. **Building Templates in AWS CloudFormaton**
        i. Review of AWS CloudFormation
        ii. AWS CloudFormation Templates
    b. Lab: Implementing AWS CloudFormation Intrinsic Functions and Keywords
    c. Knowledge Check

Hands-on labs

- **Lab: Deploying Infrastructure as Code with AWS CodePipeline (60 mins)**
  In this lab, you will use CodePipeline to deploy infrastructure as code (IaC) using an AWS CloudFormation template. This lab includes steps for building out a CloudFormation template, creating a CodePipeline by pulling source code from CodeCommit, and finally, creating and updating a CloudFormation stack.
- **Lab: Building a Serverless API with Infrastructure Composer (60 mins)**
  In this lab, you will gain hands-on experience using AWS Application Composer by creating a serverless application
- **Lab: Implementing AWS CloudFormation Intrinsic Functions and Keywords (60 mins)**
  In this lab, you will work with an existing AWS CloudFormation template that has been stripped of its intrinsic functions, conditions, and rules. The template includes comments and instructions guiding you on how to complete the missing elements. By filling in the necessary CloudFormation utilities, you will develop your understanding of these advanced features.

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Week 5: Infrastructure as Code – Part 2

Goal

Learn how to build in AWS CloudFormation, work with CloudFormation stacks, explore CloudFormation Registry, use the AWS Cloud Development Kit (AWS CDK), and test AWS CDK constructs.

Learning objectives

- Define a stack and change set
- List stack failure options in CloudFormation
- Identify the types of resources that support drift detection
- State the reasons for exporting and importing values from a stack
- Explain nested stacks
- State ways of working with Windows-based stacks
- Discuss rolling back your infrastructure and the purpose of preparing for failure

Outline

1. **Working with AWS CloudFormation Stacks**
    a. **Working with AWS CloudFormation Stacks**
        i. Building and Managing Stacks
        ii. Remediating Stack Drift
        iii. Encountering Stack Failures
        iv. Deploying Multiple Stacks
        v. Lab: Deploying Advanced CloudFormation Stacks
        vi. Knowledge Check
2. **Expanding AWS CloudFormation Usage**
    a. **Expanding AWS CloudFormation Usage**
        i. Using the CloudFormation Registry
        ii. Exploring Amazon EventBridge
        iii. AWS SimuLearn: Automation with CloudFormation
        iv. Knowledge Check
3. **Examining AWS Cloud Development Kit**
    a. **Examining AWS Cloud Development Kit**
        i. Provisioning Resources with AWS CDK
        ii. Lab: Working with Constructs in AWS CDK
        iii. Knowledge Check
4. **Testing AWS CDK Constructs**
    a. **Testing AWS CDK Constructs**
        i. Testing Methods Used with CDK Apps
        ii. Knowledge check

Hands-on labs

- **Lab: Deploying Advanced CloudFormation Stacks (60 Mins)**
  In this lab, you will use CloudFormation to create a nested stack, experience provisioning failure with stack failure options, and modify a nested stack from a change set.
- **AWS SimuLearn Challenge: Automation with CloudFormation (60 min)**
  In this lab, you will help reduce human error and standardize a robotics research deployment infrastructure.

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

- **Lab: Working with Constructs in AWS CDK (60 Mins)**
  In this lab, you will use AWS CDK within the AWS Cloud9 IDE to deploy and test a Lambda function and the API Gateway REST API.

Week 6: Logging and Scaling – Part 1

Goal

Gain knowledge of the purpose and benefits of logs. Identify AWS logging and monitoring services and learn how to apply them to optimize infrastructure management and performance.

Learning objectives

- Describe how auto scaling works
- Discuss auto scaling with Amazon ECS, Amazon DynamoDB, Amazon Aurora, and Amazon EC2
- Describe different scaling policies
- Identify necessary services to auto scale through case studies
- Describe launch templates for Amazon EC2 Auto Scaling
- Explain how automatic scaling works for Amazon EC2 Spot Fleet
- Explain how AWS Elastic Beanstalk works with an Amazon EC2 Auto Scaling group
- Explain how to use lifecycle hooks and AWS Lambda functions to create fully managed auto scaling
- Discuss predictive scaling for Amazon EC2 with machine learning
- Explain how Elastic Load Balancing (ELB) works and identify the types of ELB load balancers
- Discuss deployment strategies on AWS and identify the type of deployment by its characteristics
- Describe how to use load balancers for blue/green and canary deployments
- Discuss how ELB works with Amazon Route 53
- Describe how to use AWS PrivateLink to connect ELB services

Outline

1. **Auto Scaling and Elastic Load Balancing**
   a. **Auto Scaling**
      i. How Auto Scaling Works
      ii. Auto Scaling with Amazon ECS
      iii. Auto Scaling with DynamoDB
      iv. Auto Scaling with Aurora
      v. Amazon EC2 Auto Scaling
      vi. Automatic and Manual Scaling
      vii. Lab: Introduction to Amazon EC2 Auto Scaling
      viii. Auto Scaling Case Studies
      ix. Knowledge Check
   b. **Amazon EC2 Auto Scaling Optimization**
      i. Launch Templates
      ii. Spot Fleets and Auto Scaling
      iii. Amazon EC2 Auto Scaling Group for Elastic Beanstalk
      iv. Creating Fully Managed Auto Scaling with Lifecycle Hooks
      v. Predictive Scaling for Amazon EC2 Auto Scaling with Machine Learning
      vi. Knowledge Check

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

c. **Elastic Load Balancing**
   i. How Elastic Load Balancing Works
   ii. Elastic Load Balancing Services
   iii. Availability Zones and Load Balancer Nodes
   iv. Deployment Strategies
   v. Activity: Identifying the Type of Deployment by its Characteristics
   vi. Using an Application Load Balancer for Canary and Blue/Green Deployments
   vii. Lab: Introduction to Elastic Load Balancing
   viii. How Elastic Load Balancing Integrates with Route 53
   ix. Using PrivateLink to Connect with Elastic Load Balancing Services
   x. AWS SimuLearn: Highly Available Web Applications
   xi. Knowledge Check

Hands-on labs

- **Lab: Introduction to Amazon EC2 Auto Scaling (60 min)**
  In this lab, you will create a launch template that defines your Amazon Elastic Compute Cloud (Amazon EC2) instances and an Amazon EC2 Auto Scaling group with a single instance in it. You will then terminate the instance and verify that the instance was removed from service and replaced. To maintain a constant number of instances, Amazon EC2 Auto Scaling automatically detects and responds to Amazon EC2 health and reachability checks.
- **Lab: Introduction to Elastic Load Balancing (60 min)**
  In this lab you will create, configure, and test a network load balancer and an application load balancer.
- **AWS SimuLearn Lab: Highly Available Web Applications (60 min)**
  In this lab, you will help a travel agency create a highly available web application architecture.

Week 7: Logging and Scaling – Part 2

Goal

Learn about the purpose and benefits of logs. Identify AWS logging and monitoring services and learn how to apply them to optimize infrastructure management and performance.

Learning objectives

- Describe types of logs and the importance of logging
- Describe the monitoring features of Amazon CloudWatch
- Describe the benefits of creating a monitoring plan and defining monitoring goals
- Compare two ways of monitoring instances: using the Amazon EC2 console and using the CloudWatch dashboard
- Describe the benefits of monitoring with AWS CloudTrail
- Describe CloudWatch features for monitoring and troubleshooting
- Describe AWS X-Ray and how X-Ray works with CloudWatch
- Define high-resolution metrics and composite alarms and describe how CloudWatch uses alarms to invoke scaling

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Outline

1. **Monitoring and Amazon CloudWatch**
    a. **Logging and Monitoring in AWS**
        i. Logging Overview
        ii. Monitoring with CloudWatch
        iii. Determining a Monitoring Strategy
        iv. Establishing a Baseline with Amazon EC2 Metrics
        v. Monitoring with the CloudWatch Agent
        vi. Dashboards for Monitoring EC2 Instances
        vii. Logging Activities with CloudTrail
        viii. Knowledge Check
    b. **Monitoring Application Health**
        i. Keeping Your Applications Healthy
        ii. Serverless Observability
        iii. CloudWatch Container Insights
        iv. CloudWatch Application Insights
        v. CloudWatch Anomaly Detection
        vi. CloudWatch Metric Streams
        vii. AWS X-Ray
        viii. Knowledge Check
    c. **Amazon CloudWatch for Health, Availability, and Security**
        i. CloudWatch Dashboards
        ii. CloudWatch Synthetics and CloudWatch RUM
        iii. CloudWatch Logs Insights
        iv. CloudWatch Metrics Insights
        v. Lab: Collecting and Analyzing Logs with Amazon CloudWatch Logs Insights
        vi. CloudWatch Contributor Insights for Security
        vii. Amazon CloudWatch Evidently
        viii. Lab: Monitoring Security with Amazon CloudWatch
        ix. High-Resolution Metrics and Alarms and Composite Alarms
        x. Scaling with CloudWatch Alarms
        xi. Knowledge Check

Hands-on labs

- **Lab: Collecting and Analyzing Logs with Amazon CloudWatch Logs Insights (45 min)**
  In this lab, you will learn to use Amazon CloudWatch Logs Insights to interactively search and analyze log data in Amazon CloudWatch Logs. You will set up log groups and log streams in CloudWatch Logs and run queries against Amazon Virtual Private Cloud (Amazon VPC) flow logs and database logs to detect potential security vulnerabilities.
- **Lab: Monitoring Security with Amazon CloudWatch (45 min)**
  In this lab, you will monitor security on the database server with Amazon CloudWatch. You will configure the database server to send log files to Amazon CloudWatch. You will then create CloudWatch alarms and notifications to alert you to a specified number of login failures on your database server. Finally, you will create a CloudWatch alarm and notification to monitor outgoing traffic through a NAT gateway.

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Week 8: Monitoring and Troubleshooting – Part 1

Goal

Learn the methodology to troubleshoot common technical use cases by using AWS CloudTrail and AWS Identity and Access Management (IAM).

Learning objectives

- Explain the importance of a structured troubleshooting methodology and its benefits over ad-hoc troubleshooting approaches
- Describe the key steps involved in the troubleshooting methodology
- Apply a troubleshooting methodology to diagnose common issues related to cloud infrastructure
- Develop a troubleshooting plan for a given cloud application scenario
- Understand the purpose of AWS CloudTrail and differentiate between event types
- Analyze CloudTrail log files to identify potential security incidents
- Integrate CloudTrail with AWS services to automate log management and analysis processes
- Implement best practices for securing and managing CloudTrail trails
- Explain the purpose and functionality of CloudTrail Lake

Outline

1. **Troubleshooting with AWS CloudTrail and Amazon IAM**
    a. **Troubleshooting Methodology**
        i. Troubleshooting Methodology and Structured Approach
        ii. Using Troubleshooting Methodology in Cloud Infrastructure
        iii. Activity: Developing a Troubleshooting Plan
        iv. Knowledge Check
    b. **Troubleshooting with CloudTrail**
        i. Activity: CloudTrail Review
        ii. Analyzing AWS CloudTrail Log Files for Potential Security Incidents
        iii. CloudTrail Integration
        iv. Best Practices for Securing and Managing AWS CloudTrail
        v. CloudTrail Lake
        vi. Lab: Monitoring and Troubleshooting Amazon EC2 Workloads with Detective Controls
        vii. Knowledge Check
    c. **Troubleshooting in Amazon IAM**
        i. Activity: Amazon IAM Policies
        ii. Troubleshooting Common Errors in IAM Policies
        iii. Troubleshooting Issues in IAM Role Assumptions, Trust Relationships, and Service-Linked Roles
        iv. Activity: Tools and Services for Troubleshooting and Auditing Amazon IAM
        v. Best Practices for IAM Policy Management
        vi. Lab: Performing a Basic Audit of Your AWS Environment
        vii. Knowledge Check

Hands-on labs

- **Lab: Monitoring and Troubleshooting Amazon EC2 Workloads with Detective Controls (75 min)**
  In this lab, you will explore monitoring tools for your Amazon EC2 workloads, and apply troubleshooting steps to correct issues affecting your current workloads.

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

- **Lab: Performing a Basic Audit of Your AWS Environment (45 min)**
  In this lab, you will perform basic audits of core AWS resources. You will use the AWS Management Console to understand how to audit the use of multiple AWS services, such as Amazon EC2, Amazon VPC, Amazon IAM, Amazon Security Groups, AWS CloudTrail, and Amazon CloudWatch.

Week 9: Monitoring and Troubleshooting – Part 2

Goal

Learn the importance of metric monitoring using Amazon CloudWatch, Lambda Insights, Internet Monitor, and Network Monitor.

Learning objectives

- Use custom metrics for deeper insights
- Discuss enabling cross-account, cross-Region monitoring
- Recall CloudWatch anomaly detection and create a CloudWatch alarm based on anomaly detection
- Use CloudWatch Metrics Insights to create alarms
- Recall CloudWatch Container Insights and set up Container Insights on Amazon Elastic Container Service (Amazon ECS) and Amazon Elastic Kubernetes Service (Amazon EKS)
- Discuss troubleshooting with Container Insights
- Recall CloudWatch Lambda Insights and use Lambda Insights
- Identify root cause monitored by Lambda Insights
- Describe CloudWatch Internet Monitor and CloudWatch Network Monitor

Outline

1. **Metrics and Network Monitoring Troubleshooting**
   a. **Metric Monitoring**
      i. Custom Metrics
      ii. Activity: Data Aggregation
      iii. Cross-Account Cross-Region Monitoring
      iv. Amazon CloudWatch Anomaly Detection
      v. Alarms with Amazon CloudWatch Metrics Insights
      vi. Activity: Exploring a Use Case Scenario with Amazon CloudWatch Custom Metrics
      vii. Knowledge Check
   b. Insights and Network Monitoring
      i. AWS SimuLearn: Monitor and Analyze Network Traffic
      ii. Using Amazon CloudWatch Container Insights
      iii. Troubleshooting with CloudWatch Container Insights
      iv. Using Amazon CloudWatch Lambda Insights
      v. Troubleshooting with CloudWatch Lambda Insights
      vi. Amazon CloudWatch Internet Monitor
      vii. Activity: Amazon CloudWatch Network Monitor
      viii. AWS SimuLearn: Traffic Mirroring
      ix. Knowledge Check

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Hands-on labs

- **AWS SimuLearn Lab: Monitor and Analyze Network Traffic (60 min)**
  In this lab, you will help aerospace engineers who want to monitor and analyze network traffic to detect misuse of a space station's data and systems. The engineers want to capture information about IP traffic going to and from the network interfaces of each system, and then analyze the information to detect unusual network activity.
- **AWS SimuLearn Lab: Traffic Mirroring (60 min)**
  In this lab, the general of a galactic security fleet needs your help to ensure that their spaceships and planetary bases are secure from any potential threats. The general wants a solution to monitor the fleet's network traffic from their spaceships and bases, and test the network's effectiveness.

Week 10: Monitoring and Troubleshooting – Part 3

Goal

Learn how to implement metric filters and Live Tail for monitoring, use Amazon CloudWatch Logs, discuss Amazon CloudWatch Logs, and discuss troubleshooting with VPC Flow Logs.

Learning objectives

- Implement metric filters and Live Tail for monitoring
- Use CloudWatch Logs Insights for querying and analyzing log data
- Discuss CloudWatch Logs anomaly detection and pattern analysis
- Discuss troubleshooting with VPC Flow Logs

Outline

1. **Logs and Application Monitoring**
   a. **Logs**
      i. Metric Filters
      ii. Live Tail Data with CloudWatch Logs
      iii. Querying with CloudWatch Log Insights
      iv. Log Anomaly Detection and Pattern Analysis
      v. Troubleshooting with VPC Flow Logs
      vi. Lab: Monitoring Applications and Infrastructure
      vii. Knowledge Check
   b. Application Monitoring
      i. Using CloudWatch Application Insights
      ii. Using CloudWatch Synthetics
      iii. Using CloudWatch RUM
      iv. Application Signals
      v. Knowledge Check

Hands-on labs

- **Lab: Monitoring Applications and Infrastructure (60 Mins)**
  In this lab, you will practice setting up and monitoring metrics for business-application events, define relevant event thresholds for metrics, create an automated notification, and perform a remediation when metric thresholds are exceeded.

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Week 11: Monitoring and Troubleshooting – Part 4

Goal

Build foundational knowledge about the principles and implementations of distributed tracing with AWS X-Ray.

Learning objectives

- Discuss the features, benefits, and concepts of AWS X-Ray
- Explain traces
- Explore service graphs
- Explore trace maps
- Describe how to use filter expressions
- Describe cross-account tracing
- Understand how to use tracing with X-Ray
- Explain latency histograms
- Discuss X-Ray insights
- Describe the X-Ray Analytics console
- Discuss X-Ray SDK for Python

Outline

1. **Tracing**
    a. **VPC Reachability Analyzer**
        i. Network Troubleshooting
        ii. Lab: Troubleshooting Website Reachability Behind a Load Balancer
        iii. Reachability Analyzer Functionality and Use Cases
        iv. Activity: Troubleshooting with Reachability Analyzer
        v. Knowledge Check
    b. **Tracing with AWS X-Ray**
        i. X-Ray Features and Functionality
        ii. X-Ray Concepts
        iii. Tracing an Application
        iv. Knowledge Check
    c. **Troubleshooting and Monitoring with AWS X-Ray**
        i. Activity: Latency Histograms
        ii. Activity: X-Ray Insights and Analytics
        iii. X-Ray SDK for Python
        iv. AWS SimuLearn: Core Security Concepts
        v. Knowledge Check

Hands-on labs

- **Lab: Troubleshooting Website Reachability Behind a Load Balancer (60 Mins)**
  In this lab, you will explore the concepts of configuring a web server behind an Elastic Load Balancer (ELB). You will address a scenario where users are not able to reach a website hosted on an Amazon Elastic Compute Cloud (Amazon EC2) instance behind a load balancer.
- **AWS SimuLearn Lab: Core Security Concepts (60 min)**
  In this lab, you will improve security at the city's stock exchange by ensuring that support engineers can only perform authorized actions.

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.